Hybrid Mode Integration Guide

API Contract & Optimization Process

Traigent SDK v1.0

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Confidential — for integration partners only

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Overview

Traigent Hybrid Mode enables zero-code optimization of external agentic services. Your service exposes a lightweight REST API; Traigent's optimizer discovers tunable parameters, explores configurations, and converges on an optimal setting—all without accessing your underlying data.

What you implement

A REST service with the prefix `/traigent/v1/` exposing the endpoints listed in Section 5. Three endpoints are required; three are optional.

Required endpoints: `GET /traigent/v1/capabilities`, `GET /traigent/v1/config-space`, and `POST /traigent/v1/execute`.

Optional endpoints: `POST /traigent/v1/evaluate`, `GET /traigent/v1/health`, and `POST /traigent/v1/kee`

What Traigent provides

- Automatic parameter-space exploration (NSGA-II, Bayesian, grid).
- Multi-objective optimization (accuracy, cost, latency, or any custom metric).
- Budget controls (max trials, max spend, wall-clock limits).
- Statistical promotion via ε-Pareto dominance testing.

Integration readiness checklist

Before production rollout, validate that:

- `GET /traigent/v1/capabilities` returns `version` and accurate feature flags.
- `GET /traigent/v1/config-space` returns a stable `capability_id` and valid `tunables`.
- `POST /traigent/v1/execute` enforces capability matching and always returns `operational_metrics.tot`
- `POST /traigent/v1/evaluate` is implemented whenever `supports_evaluate=true`.
- Optional sections (`constraints`, `objectives`, `exploration`, `promotion_policy`, `defaults`, `measures`) are omitted unless intentionally used.

Optimization Process

The optimization loop follows four phases, repeated until a budget or convergence criterion is met.

1. **Discover** — Traigent calls `GET /traigent/v1/capabilities` and `GET /traigent/v1/config-space` to learn which parameters are tunable, their domains, and declared objectives.
2. **Execute** — For each trial, Traigent posts a candidate configuration and a batch of inputs to `POST /traigent/v1/execute`. Your service runs the agent with those settings and returns per-input outputs plus operational metrics (cost, latency, tokens).
3. **Evaluate** — If a separate evaluation endpoint is available, Traigent posts output–target pairs to `POST /traigent/v1/evaluate`. Your service scores outputs and returns quality metrics (accuracy, relevance, etc.).
4. **Optimize** — Traigent aggregates metrics, updates its surrogate model, and selects the next configuration to try. After all trials complete, the optimizer ranks configurations by ε-Pareto dominance and reports the best.

Phase	Direction	Endpoint
Discover	Traigent → Service	`GET /traigent/v1/capabilities`, `GET /traigent/v1/config-space`
Execute	Traigent → Service	`POST /traigent/v1/execute`
Evaluate	Traigent → Service	`POST /traigent/v1/evaluate`
Optimize	Internal	(no API call)

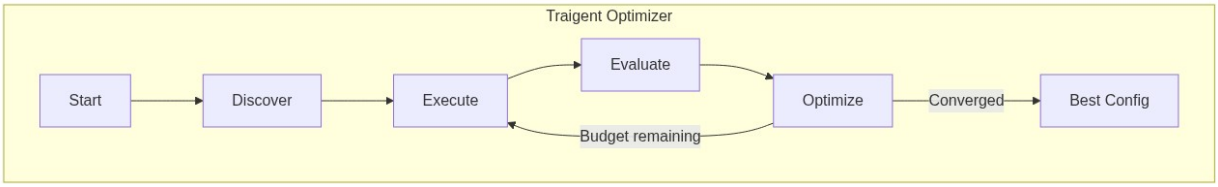

Figure 1: Traigent optimization loop: the optimizer iterates through Execute and Evaluate phases until the budget is exhausted or convergence is reached.

Operational timing guidance

Use the following targets so clients can set timeout and retry behavior predictably:

Endpoint	Target latency	Guidance
`GET /traigent/v1/capabilities`	p95 < 1s	Keep lightweight and cacheable.
`GET /traigent/v1/config-space`	p95 < 1s	Should be deterministic and idempotent.
`POST /traigent/v1/execute`	p95 < `timeout_ms`	Default timeout budget is 30s; wrapper returns HTTP 408 when exceeded.
`POST /traigent/v1/evaluate`	p95 < `timeout_ms` (if provided)	Optional timeout budget; otherwise governed by client transport timeout.
`POST /traigent/v1/keep-alive`	p95 < 500ms	Heartbeat path should remain fast and side-effect constrained.

Transport note: production integrations should use `HTTPS + HTTP/2`. Traigent's HTTP client can enforce this in strict mode (`create_transport(..., require_http2=True)`), which rejects non-HTTPS endpoints and non-HTTP/2 responses. The v1 contract remains synchronous request/response. Async job-style APIs and gRPC are future roadmap options, not part of this contract version.

Privacy-Preserving Mode

Privacy is the default. Only configuration values and numeric metrics are observed by Traigent. Input data, output content, and evaluation targets are **never** transmitted.

Data	Observed by Traigent
Tunable definitions (config-space)	Yes
Configuration values per trial	Yes
Operational metrics (cost, latency, tokens)	Yes
Quality metrics (accuracy, relevance, etc.)	Yes
Input data content	**No** (only IDs)
Output data content	**No** (only IDs)
Target / expected output content	**No** (only IDs)

Your service stores data locally and uses opaque identifiers (`input_id`, `output_id`, `target_id`) in API exchanges. If privacy is not a concern, you may include full content in `data`, `output`, and `target` fields. Both modes can be mixed.

Authentication (Optional)

If your service requires authentication, accept a standard `Authorization` header (for example a bearer token). Configure the Traigent optimizer with:

```
@traigent.optimize(
    execution_mode="hybrid_api",
    hybrid_api_endpoint="https://api.your-service.com",
    capability_id="my_agent",
    hybrid_api_auth_header="Bearer <token>",
)
```

All endpoints must consistently enforce the same authentication policy. In OpenAPI, this is modelled as optional bearer auth (`bearerAuth` or anonymous).

Roadmap: stronger policy layers (for example RBAC/tenant-scoped roles) can be added later without changing endpoint shapes.

Endpoints

All paths are prefixed with `/traigent/v1/`.

GET /traigent/v1/capabilities *(required)*

Returns service capabilities for the initial handshake.

Response (200)

```
{
  "version": "1.0",
  "supports_evaluate": true,
  "supports_keep_alive": false,
  "supports_streaming": false,
  "max_batch_size": 100,
  "max_payload_bytes": null,
  "capability_ids": ["my_agent"]
}
```

Guidance: omit optional top-level sections when they are not used. Avoid sending placeholder empty values unless they carry intent.

Field	Type	Req.	Description
`version`	string	**Yes**	API version (currently `"1.0"`)
`supports_evaluate`	boolean	**No**	Separate `/traigent/v1/evaluate` endpoint available (default: `true`)
`supports_keep_alive`	boolean	**No**	Session heartbeat supported
`supports_streaming`	boolean	**No**	Streaming responses supported
`max_batch_size`	integer	**No**	Max inputs per execute request (default: 100)
`max_payload_bytes`	integer	**No**	Max request payload size (`null` = unlimited)
`capability_ids`	array[string]	**No**	Optional list of capability IDs supported by this service

Important: discovery responses do **not** return `config_id` / `config_ids`. The canonical capability identifier for execution/evaluation is `capability_id` from `GET /traigent/v1/config-space`. Per-run identifiers are returned later by `POST /traigent/v1/execute` as `execution_id`.

Note: `capability_ids` in `/capabilities` is optional and primarily useful for service introspection. The canonical capability used for execution/evaluation is `capability_id` returned by `GET /traigent/v1/config-space`.

GET /traigent/v1/config-space *(required)*

Returns tunable variable definitions. Traigent uses these to build the parameter search space.

Response (200)

```
{
  "schema_version": "0.9",
  "capability_id": "my_agent",
  "tunables": [
    {
      "name": "model",
      "type": "enum",
      "domain": {"values": ["gpt-4o", "claude-3-sonnet", "llama-70b"]},
      "default": "gpt-4o"
    },
    {
      "name": "temperature",
      "type": "float",
      "domain": {"range": [0.0, 2.0], "resolution": 0.1},
      "default": 0.7
    }
  ],
  "objectives": [
    {"name": "accuracy", "direction": "maximize", "weight": 2.0},
    {"name": "cost",     "direction": "minimize", "weight": 1.0}
  ]
}
```

Field	Type	Req.	Description
`schema_version`	string	**Yes**	TVL schema version (currently `"0.9"`)
`capability_id`	string	**Yes**	Unique identifier for this capability
`tunables`	array	**Yes**	List of tunable definitions (see below)
`constraints`	object | array	**No**	Structural / behavioral constraints
`objectives`	array	**No**	Objective definitions (name, direction, weight)
`exploration`	object	**No**	Strategy, budgets, and convergence criteria
`promotion_policy`	object	**No**	ε-Pareto promotion policy
`defaults`	object	**No**	Default configuration values
`measures`	array	**No**	Declared metric names produced by the service

Tunable Definition

Each element of the `tunables` array describes one parameter:

Field	Type	Req.	Description
`name`	string	**Yes**	Variable name (valid Python identifier)
`type`	string	**Yes**	One of: `enum`, `float`, `int`, `bool`, `str`
`domain`	object	Conditional	Domain specification (required for all types except `bool`)
`default`	any	**No**	Default value
`agent`	string	**No**	Agent name for multi-agent grouping

Tunable Types

Type	Domain format	Example
`enum`	`{"values": [...]}`	`["gpt-4o", "claude-3-sonnet"]`
`float`	`{"range": [min, max], "resolution": step}`	`[0.0, 2.0]`, step 0.1
`int`	`{"range": [min, max]}`	`[100, 4096]`
`bool`	N/A	`true` / `false`
`str`	`{"values": [...]}`	`["concise", "detailed"]`

POST /traigent/v1/execute *(required)*

Execute the agent with a specific configuration on a batch of inputs.

Request

```
{
  "request_id": "550e8400-e29b-41d4-a716-446655440000",
  "capability_id": "my_agent",
  "config": {
    "model": "gpt-4o",
    "temperature": 0.5,
    "max_tokens": 1024
  },
  "inputs": [
    {"input_id": "ex_001", "data": {"query": "What is machine learning?
        "}},
    {"input_id": "ex_002", "data": {"query": "Explain neural networks"
        }}
  ],
  "timeout_ms": 30000
}
```

Field	Type	Req.	Description
`request_id`	string (UUID)	**No**	Idempotency key (auto-generated if omitted)
`capability_id`	string	**Yes**	Capability to invoke
`config`	object	**Yes**	Configuration parameters (tunable values)
`inputs`	array	**Yes**	Input examples to process
`session_id`	string	**No**	Session ID for stateful agents
`batch_options`	object	**No**	Parallelism, fail-fast, per-item timeout
`timeout_ms`	integer	**No**	Request timeout in ms (default: 30 000)

Each **Input Item** contains:

`input_id`	string	**Yes**	Unique identifier for this input
`data`	object	**No**	Input data (omit in privacy-preserving mode)

Response (200)

```
{
  "request_id": "550e8400-e29b-41d4-a716-446655440000",
  "execution_id": "exec_123456",
  "status": "completed",
  "outputs": [
    {
      "input_id": "ex_001",
      "output": {"response": "Machine learning is a subset of AI..."},
      "cost_usd": 0.002,
      "latency_ms": 150
    }
  ],
  "operational_metrics": {
    "total_cost_usd": 0.005,
    "latency_ms": 330,
    "tokens_used": 450
  }
}
```

Field	Type	Req.	Description
`request_id`	string	**Yes**	Echoed request ID for correlation
`execution_id`	string	**Yes**	Unique execution ID (referenced by `/traigent/v1/evaluate`)
`status`	string	**Yes**	`"completed"`, `"partial"`, or `"failed"`
`outputs`	array	**Yes**	Per-input results (see Output Item below)
`operational_metrics`	object	**Yes**	Aggregate operational metrics
`quality_metrics`	object	**No**	Quality metrics (combined execute+evaluate mode)
`session_id`	string	**No**	Session ID for stateful agents
`error`	object	**No**	Error details if `status` is `"failed"` or `"partial"`

Each **Output Item** contains:

`input_id`	string	**Yes**	Matching input identifier
`output`	object	**No**	Agent output (omit in privacy-preserving mode)
`output_id`	string	**No**	Output identifier (privacy-preserving mode)
`cost_usd`	number	**No**	Cost for this input
`latency_ms`	number	**No**	Latency for this input
`metrics`	object	**No**	Per-input quality metrics (combined mode)
`error`	string	**No**	Error message for partial failures

Operational Metrics

`total_cost_usd` is the only required field. Additional recommended and custom metrics may be included:

Field	Type	Description
`total_cost_usd`	number	**Required.** Total cost in USD
`latency_ms`	number	Recommended. Total latency in milliseconds
`tokens_input`	integer	Input token count
`tokens_output`	integer	Output token count
`cache_hit_rate`	number	Cache hit ratio (0.0–1.0)
`retry_count`	integer	Number of retries

Any additional numeric, string, or boolean fields are accepted.

POST /traigent/v1/evaluate *(optional)*

Evaluate outputs against expected targets. Only required when `supports_evaluate` is `true` in capabilities.

Request

```
{
  "request_id": "660e8400-e29b-41d4-a716-446655440001",
  "capability_id": "my_agent",
  "execution_id": "exec_123456",
  "evaluations": [
    {
      "input_id": "ex_001",
      "output": {"response": "Machine learning is a subset of AI..."},
      "target": {"expected": "Machine learning is a branch of AI..."}
    }
  ],
  "timeout_ms": 30000
}
```

Field	Type	Req.	Description
`request_id`	string (UUID)	**No**	Idempotency key
`capability_id`	string	**Yes**	Evaluation capability identifier
`execution_id`	string	**No**	Reference to previous execute
`evaluations`	array	**No**	Output–target pairs (typically non-empty)
`config`	object	**No**	Evaluation parameters
`session_id`	string	**No**	Session ID for stateful agents
`timeout_ms`	integer	**No**	Optional timeout budget in ms (wrapper emits 408 on timeout)

Note: `capability_id` is validated against the configured service capability.

Each **Evaluation Item** contains:

`input_id`	string	**Yes**	Matching input identifier
`output`	object	**No**	Agent output (or use `output_id`)
`output_id`	string	**No**	Output reference (privacy-preserving mode)
`target`	object	**No**	Expected output (or use `target_id`)
`target_id`	string	**No**	Target reference (privacy-preserving mode)

Response (200)

```
{
  "request_id": "660e8400-e29b-41d4-a716-446655440001",
  "status": "completed",
  "results": [
    {
      "input_id": "ex_001",
      "metrics": {"accuracy": 0.92, "relevance": 0.88, "fluency": 0.95}
    }
  ],
  "aggregate_metrics": {
    "accuracy":  {"mean": 0.92, "std": 0.05, "n": 100},
    "relevance": {"mean": 0.88, "std": 0.03, "n": 100}
  }
}
```

Field	Type	Req.	Description
`request_id`	string	**Yes**	Echoed request ID
`status`	string	**Yes**	`"completed"`, `"partial"`, or `"failed"`
`results`	array	**Yes**	Per-example evaluation results
`aggregate_metrics`	object	**Yes**	Aggregated statistics per metric
`error`	object	**No**	Error details for partial/failed evaluations

Each **Result Item**: `input_id` (required), `metrics` (required object of numeric quality scores), `error` (optional string).

Each **Aggregate Metric**: `mean` (number), `std` (number), `n` (integer).

GET /traigent/v1/health *(optional)*

Health check for monitoring and load-balancer probes.

Field	Type	Req.	Description
`status`	string	**Yes**	`"healthy"`, `"degraded"`, or `"unhealthy"`
`version`	string	**No**	Service version
`uptime_seconds`	number	**No**	Uptime in seconds
`details`	object	**No**	Additional health details

POST /traigent/v1/keep-alive *(optional)*

Session heartbeat for stateful agents. Only implement if `supports_keep_alive` is `true` in capabilities.

Request: `{"session_id": "session_abc123"}`

200 Response: `{"status": "alive", "session_id": "session_abc123"}`

404 Response: `{"error": {"code": "SESSION_NOT_FOUND", "message": "..."}}`

Error Responses

All endpoints must return structured errors with appropriate HTTP status codes.

Status	Description	Client behavior
400	Bad Request (invalid input, capability mismatch)	Do not retry until payload is fixed.
401	Unauthorized (when endpoint authentication is enabled)	Refresh/reconfigure credentials, then retry.
404	Not Found (unknown route or session)	Re-discover endpoint/state before retry.
408	Request Timeout (`timeout_ms` exceeded)	Retry with backoff and/or larger timeout budget.
429	Too Many Requests (rate limited)	Retry with backoff; respect `Retry-After`.
500	Internal Server Error	Retry with bounded backoff and alert if persistent.
503	Service Unavailable	Retry with backoff; respect `Retry-After`.

Note: endpoint-specific response sets are defined in `hybrid-mode-openapi.yaml` and are the normative source.

Error body format:

```
{
  "error": {
    "code": "INVALID_CONFIG",
    "message": "Unknown tunable: invalid_param",
    "details": {"invalid_params": ["invalid_param"]}
  }
}
```

Observability and trace context

For distributed tracing, clients may send W3C headers: `traceparent` and `tracestate`. Wrapper responses echo these headers and include `x-traigent-request-id` when a request/response contains `request_id`. Keep `request_id` for business-level correlation and trace headers for span-level correlation.

Idempotency

`GET /traigent/v1/capabilities`, `GET /traigent/v1/config-space`, and `GET /traigent/v1/health` should be side-effect free and idempotent by contract.

`POST /traigent/v1/execute` and `POST /traigent/v1/evaluate` accept `request_id` (UUID recommended). If the same `request_id` is received again, the service may return a cached response. This enables safe retries without duplicate processing.

Required server behavior for idempotency keys:

- Same `request_id` + same payload → return cached response.
- Same `request_id` + different payload → return HTTP 400.

The `POST /traigent/v1/keep-alive` endpoint uses `session_id` and does not require `request_id`.

Wrapper note: `TraigentService` implements in-process idempotency for `/execute` and `/evaluate`. If custom servers do not implement idempotency, retrying the same request can execute work twice and duplicate side effects/cost.

Optional extension: `OPTIONS` can be implemented for method discovery/CORS convenience, but is not required by the core hybrid contract.

Accompanying Specifications

The following machine-readable files are provided alongside this document and constitute the normative contract:

File	Format	Usage
`hybrid-mode-openapi.yaml`	OpenAPI 3.0.3	Client/server codegen, request/response validation, Swagger UI
`hybrid-mode-schemas.json`	JSON Schema (Draft 2020-12)	Standalone DTO validation
`hybrid-mode-api-contract.md`	Markdown	Full endpoint reference with examples
`hybrid-mode-client-guide.md`	Markdown	Implementation patterns, code samples, troubleshooting

Generating client stubs

```
// Python
openapi-generator generate -i hybrid-mode-openapi.yaml -g python -o client/

// TypeScript
openapi-generator generate -i hybrid-mode-openapi.yaml -g typescript-fetch -o
    client/
```

Validation and linting

```
# Validate syntax and references
npx @apidevtools/swagger-cli validate hybrid-mode-openapi.yaml

# Lint for OpenAPI best practices
npx @redocly/cli lint hybrid-mode-openapi.yaml
```

Publish as single source of truth

Publish `hybrid-mode-openapi.yaml` in Swagger UI/Redoc from versioned git tags (for example `v1.0.0`, `v1.0.1`). Avoid distributing hand-edited zip snapshots as the primary contract source.

Quick Reference

Method	Path	Req.	Purpose
GET	/traigent/v1/capabilities	**Yes**	Service handshake
GET	/traigent/v1/config-space	**Yes**	Tunable definitions
POST	/traigent/v1/execute	**Yes**	Run agent with config
POST	/traigent/v1/evaluate	**No**	Score outputs vs. targets
GET	/traigent/v1/health	**No**	Health check
POST	/traigent/v1/keep-alive	**No**	Session heartbeat